

Course Title	PRODUCT DEVELOPMENT
Course Code	ME625
Course Credit	Theory :03
	Practical :01
	Tutorial :00
	Credits :04

Course Learning Outcomes:

After Successful completion of the above course, students will be able to:

- **Identity** and **Implement** various prototyping methods.
- **Develop** a product and **Create** various models
- **Describe** Industrial Product Design methods.
- **Classify** 3d Printing techniques and machines
- **Use** 3D printing slicing software
- **Write** Business plans

Detailed Syllabus

Sr. No.	Name of chapter & details	Hours Allotted
SECTION-I		
1.	Introduction Difference between product design and product development. Importance for Engineering applications.	02
2.	Prototyping Need of prototyping, Analytical Prototype, Physical Prototype, Choosing right prototyping process, Prototyping methods and tools, Mock-up, Model, Prototype, Virtual Prototyping,	08
3.	Model Making Model making using waste material, Paper Model, Card board models, Thermocol models. Model making with low cost materials.	04
4.	Case studies and projects	07
Total		21

SECTION-II

5.	Industrial Design: Definition, Assessing the need of industrial design, Design in context of Business, Manufacturing Processes, Material Selection. Design for assembly, Design for Manufacturing, Design for Environment,	02
6.	3d Printing What is 3d printing ? 3D Printing technology and processes, Types of 3D printing machines, Materials, Applications, CAD to printing.	04
7.	3D printer Slicing Application – CURA Initial setup, Interface, Settings, Preview, Print.	06*
8.	Business Plan Business Proposal Writing : Product & Service Description, Opportunity, Sales Literature, Technology/Methodology, Marketing Plan, Operational Plan, Financial Plan, Future Products and Services	05
9.	Case studies and projects	10
Total		21

Instructional method and Pedagogy:

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory which carries 10 Marks weight age.
- Two internal exams will be conducted and each of the same will be converted to equivalent of 20 Marks as a part of internal theory evaluation.
- Assignments based on course content will be given to the students at the end of each unit/topic and will be evaluated at regular interval. It carries a weight age of 10 Marks as a part of internal theory evaluation.
- Surprise tests/Quizzes/Seminar will be conducted which carries 10 Marks as a part of internal theory evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.
- Experiments/Tutorials related to course content will be carried out in the practical session.
- In the lectures and laboratory discipline and behaviour will be observed strictly.

Reference Books:

5. Cao Jerry, Zieba Kamil, Ellis Matt, "The Ultimate Guide to Prototyping", UXPin Inc.
6. Cuffaro F. Daniel et.al, "The Industrial Design Reference + Specification Book",
7. Karl T. Ulrich, "Product design and development", Tata McGraw Hill.
8. Kevin Otto, Kristin Wood, "Product Design – Techniques in Reverse Engineering and New product Development", Pearson publication.
9. A C Chitale and R C Gupta, "Product Design and Manufacturing", Prentice Hall New Delhi
10. George E. Dieter and Linda C. Schmidt, "Engineering Design", Tata McGraw Hill

Additional Resources

- <https://academy.autodesk.com/course/134426/advanced-cad-manufacturing>
- <https://academy.autodesk.com/course/133268/create-drawings-manufacture-and-build-3d-printer>
- <https://www.justinmind.com/blog/5-prototyping-techniques-to-improve-the-way-you-present-your-designs/>
- <https://www.interaction-design.org/literature/article/prototyping-learn-eight-common-methods-and-best-practices>
- <http://www.dsourc.in/course/product-design-1>
- <http://www.modelmakers-uk.co.uk/>